

Review

HDAC6 inhibitors: Translating genetic and molecular insights into a therapy for axonal CMT



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HIGHLIGHTS

- Axonal transport defects are a common hallmark in models of axonal CMT.
- HDAC6 plays a central role in several pathomechanisms of axonal CMT.
- HDAC6 inhibitors are a promising class of novel therapeutics for axonal CMT.

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ABSTRACT

Histone deacetylase 6 (HDAC6) plays a central role in various processes that are key for neuronal survival. In this review, we summarize the current evidence related to disease pathways in the axonal form of Charcot-Marie-Tooth disease (CMT) and highlight the role of HDAC6 in these pathways. We hypothesize that HDAC6 might in fact actively contribute to the pathogenesis of certain forms of axonal CMT. HDAC6 plays a deacetylase activity-dependent, negative role in axonal transport and axonal regeneration, which are both processes implicated in axonal CMT. On the other hand, HDAC6 coordinates a protective response during elimination of toxic misfolded proteins, but this is mostly mediated independent of its deacetylase activity. The current mechanistic insights on these functions of HDAC6 in axonal CMT, along with the selective druggability against its deacetylase activity, make the targeting of HDAC6 particularly attractive. We elaborate on the preclinical studies that demonstrated beneficial effects of HDAC6 inhibitors in axonal CMT models and outline possible modes of action. Overall, this overview ultimately provides a rationale for the use of small-molecule HDAC6 inhibitors as a therapeutic strategy for this devastating disease.

1. Introduction

Charcot-Marie-Tooth disease (CMT) embodies a spectrum of disorders that affect the peripheral nervous system and these diseases are clinically defined by the progressive loss of force and touch sensation (Bird, 1993). Patients with CMT experience difficulty walking and frequently trip or fall because of weakness of the lower legs and feet. Later in the disease course, they can present with foot deformities such as high arches and hammertoes, along with weakness and atrophy of the hands (Bird, 1993). In many cases, also the sense of touch and temperature is diminished. The onset of clinical symptoms often occurs in the first or second decade of life and slowly progresses, generally not affecting life expectancy. CMT is the most common form of inherited neuropathies, with an estimated prevalence of 1 in 2500 individuals (Barreto et al., 2016). It is a highly heterogeneous disease, both

clinically and genetically.

CMT patients can be categorized in two major types: CMT type 1 (CMT1), which comprises neuropathies that predominantly originate from dysfunctions in the myelin sheaths surrounding the axons, and CMT type 2 (CMT2), which covers the neuropathies originating from axonal degeneration (Bird, 1993). Also intermediate forms exist, in which the patient shows signs of both demyelination and axonal degeneration (CMT type 4; CMT4). CMT type X (CMTX) is caused by mutations on the X chromosome.

So far, more than 80 genes are known to be associated with CMT (Laurá et al., 2019). Despite our expanding knowledge on the underlying genetic causes, no therapies exist that can halt or reverse the symptoms of this disorder. Symptomatic care, such as physiotherapy and therapeutic exercises are mainly used to comfort patients and to maximize functional use of all extremities (McCorquodale et al., 2016).

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As a consequence, there is a strong need for effective therapies for CMT.

Most research focusses on finding a successful therapy for CMT1A caused by a PMP22 gene duplication, as it is the underlying cause of nearly half of the CMT cases (Juneja et al., 2019). On the contrary, not many therapeutic strategies were tested for other forms of CMT, including CMT2.

Some years ago, we discovered that treatment with HDAC6 inhibitors partially rescued the CMT2 phenotype of mutant HSPB1 mice (d'Ydewalle et al., 2011). Cultured dorsal root ganglion (DRG) neurons isolated from mutant HSPB1 mice showed defective axonal transport of mitochondria and this was associated with a decrease in α -tubulin acetylation. Small molecule HDAC6 inhibitors could restore acetylated α -tubulin levels and mitochondrial transport (d'Ydewalle et al., 2011). Besides, these treatments also partially rescued the motor and sensory phenotypes in the mutant HSPB1 mice (d'Ydewalle et al., 2011). Additional work from our and other research groups extended these findings to GARS-mediated CMT2 (Benoy et al., 2018; Mo et al., 2018). Studies in iPSC-derived motor neurons from HSPB1 patients further validated the efficiency of this therapy (Kim et al., 2016), as well as a recent study using neuronal explant cultures expressing mutant HSPB1 (Adalbert et al., 2020).

In this review, we highlight several potential disease pathways underlying the axonal form of CMT. We discuss the role of HDAC6 in these pathways and propose a model on how HDAC6 might directly participate in the CMT2 pathogenesis. Furthermore, we elaborate on the current knowledge on the therapeutic efficacy of HDAC6 inhibitors in preclinical CMT2 models.

2. Genetic and molecular clues to disease pathways in axonal CMT

CMT2, the axonal form of CMT, accounts for approximately 20% of the total CMT patient population (Juneja et al., 2019). This form of the disease is caused by genetic mutations that induce direct damage to the peripheral axons. CMT2 is genetically highly heterogeneous, which could be responsible for the limited therapeutic strategies that were tested in this patient population.

Over the years, genetic and molecular research has provided unprecedented insights into the pathogenic mechanisms underlying axonal CMT. We believe that this knowledge is a prerequisite for the development of effective therapies that can slow down, halt or reverse the symptoms of this dreadful disease.

2.1. Axonal transport defects are a common phenotype of CMT2

Axonal transport in neurons is key for the delivery of essential components, including RNA, lipids, proteins and mitochondria, to the synapses and their transport back for degradation, recycling or signaling.

Pioneering work from Baloh et al. showed that mutations in *Mitofusin-2* (*MFN2*), which is the most common cause of CMT2, drive axonal transport defects of mitochondria in cultured rat DRG neurons (Baloh et al., 2007). This phenotype was later confirmed in zebrafish and mouse *MFN2* models (Chapman et al., 2013; Rocha et al., 2018). Over the years, axonal transport defects were revealed in various *in vitro* and *in vivo* models for other CMT2-causative genes, including *RAB7*, *HSPB1*, *GARS*, *KIF1B* and *KIF5A* (Benoy et al., 2018; Campbell et al., 2014; d'Ydewalle et al., 2011; Janssens et al., 2014; Kalmar et al., 2017; Kim et al., 2016; Mo et al., 2018; Ponomareva et al., 2016; Xu et al., 2018; Zhang et al., 2013). An overview of these studies is shown in Table 1.

Axonal transport deficits therefore seem to be a common molecular phenotype of CMT2 pathogenesis (Prior et al., 2017). Interestingly, it became clear that the transport of a diverse set of cargoes can become hampered, suggesting that the transport machinery is primarily affected. Nevertheless, two studies have shown that also the cargo itself, in particular the mitochondria, can become impaired by some of these

disease mutations (Kalmar et al., 2017; Rocha et al., 2018). These defects could presumably further contribute to the disease progression.

The observed perturbations in axonal transport could actually explain the selective vulnerability of peripheral neurons in CMT2 pathogenesis. These neurons have the longest axons of all neuronal subtypes and are therefore particularly dependent on proper intracellular transport to maintain normal structure and function (Peters et al., 2015).

However, two important questions still remain. Which are the molecular mechanisms underlying the transport defects and are these alterations in axonal transport a cause or a consequence of the disease?

For several CMT2 cases, there are strong indications for a causal relationship between axonal transport defects and the disease. *MFN2* is an outer mitochondrial membrane protein and it was shown to participate in Miro/Milton-mediated tethering of mitochondria to the kinesin complex (Misko et al., 2010). Therefore, it is reasonable to conclude that the observed axonal transport defects of mitochondria in mutant *MFN2* models are a direct result of its impaired function (Baloh et al., 2007; Chapman et al., 2013; Rocha et al., 2018). Similarly, *Rab7* is known to play a key role during axonal transport, in particular during retrograde transport of signaling endosomes (Deinhardt et al., 2006). As a consequence, the observed variability in the transport defects of different endosome types in mutant *Rab7* models makes sense (Janssens et al., 2014; Ponomareva et al., 2016; Zhang et al., 2013). The identification of mutations in kinesin and dynein motor genes, *KIF5A* (Nicolas et al., 2018), *KIF1B* (Zhao et al., 2001) and *DYNC1H1* (Weedon et al., 2011), in combination with the uncovering of axonal transport defects in mutant kinesin models (Campbell et al., 2014; Xu et al., 2018) further strengthens the conclusion that axonal transport defects actively contribute to CMT2 pathogenesis.

2.2. Impairment of axonal regeneration in CMT2

During neuronal development, growing axons are guided by specific extracellular cues to reach their proper targets. Also adult neurons maintain the ability to grow, which is required for plasticity and branching as well as for regeneration after injury. Axonal outgrowth progresses through different stages that are mediated by the intimate coordination of microtubules and filamentous (F)-actin (Dent et al., 2011).

It is becoming increasingly recognized that the capacity and the molecular mechanisms underlying regeneration differ between the central (CNS) and peripheral nervous system (PNS) (Ertürk et al., 2007; Huebner and Strittmatter, 2009). The main explanatory factor for these differences in regenerating capacity is the extracellular environment. After traumatic injury or neurodegeneration in the CNS, components of the fragmented myelin and of the glial scar impede axonal growth (McKerracher and Rosen, 2015; Siebert et al., 2014). In contrast, the PNS features a growth-permissive environment that is mediated by Schwann cells and macrophages (Bosse, 2012).

Nevertheless, there are indications that the process of axonal regeneration in the PNS is impaired in axonal CMT. Mutations in *Rab7* were shown to affect axonal growth in various neuronal cell types and in a zebrafish model (Cogli et al., 2010; Ponomareva et al., 2016; Yamauchi et al., 2010). In addition, mutations in *GARS* and *KIF1B* caused impairments in axonal outgrowth (Benoy et al., 2018; Xu et al., 2018). The decline in axonal outgrowth seems to be age-dependent, as outgrowth defects are observed in DRG cultures of one-year-old, but not of one-month-old, mutant *GARS* mice (Benoy et al., 2018; Sleight et al., 2017). It is plausible that disrupted axonal transport partially underlies these observations, as it interferes with the trophic signaling and energy demands after initial injury. Therefore, axonal transport defects could create a vicious cycle of increased degeneration and impaired regeneration of peripheral neurons.

Table 1

Axonal transport defects are a common hallmark in CMT2 models. Blue: decreased/slower, red: increased/faster, green: no change, grey: not studied, T: total. +: anterograde, -: retrograde, Tub-Ac: levels of α -tubulin acetylation, DRG: dorsal root ganglion neuron, HIPP: hippocampal neuron, MN: motor neuron, NGF: nerve growth factor, p75NTR: neurotrophin receptor p75, IGF1R: insulin-like growth factor 1.

CMT2	Gene	Mutation	Model	Axonal transport defect			Cargo		Tub-Ac	Ref.
				T	+	-	Type	Defect		
MFN2		V69F, L76P	1° DRG				Mitochondria			(Baloh et al., 2007)
		L285X	Zebrafish				Mitochondria			(Chapman et al., 2013)
		T105M	1° HIPP + Sciatic nerve				Mitochondria			(Rocha et al., 2018)
Rab7		Q67L, N161T	1° DRG				Rab7 endosomes			(Zhang et al., 2013)
		N161T	1° DRG				NGF endosomes			(Zhang et al., 2013)
		L129F	<i>Drosophila</i>				Rab7 endosomes			(Janssens et al., 2014)
		L129F, N161T, V162M, K157N	Zebrafish				Endosomes			(Ponomareva et al., 2016)
HSPB1		S135F, P182L	1° DRG				Mitochondria			(d'Ydewalle et al., 2011)
		S135F, P182L	iPSC-MN				Mitochondria			(Kim et al., 2016)
		S135F, P39L, R140G	1° MN				Mitochondria			(Kalmar et al., 2017)
		S135F, P39L, R140G	1° MN				p75NTR			(Kalmar et al., 2017)
GARS		C201R	1° DRG				Mitochondria			(Benoy et al., 2018)
		P234KY	1° DRG				NGF endosomes			(Mo et al., 2018)
KIF1B		Y1087C	1° HIPP				IGF1R endosomes			(Xu et al., 2018)
KIF5A		sa7168	<i>Drosophila</i>				Mitochondria			(Campbell et al., 2014)

2.3. Evidence for protein turnover deficits and protein aggregation in CMT2

A critical cellular process is the proper folding of proteins into their native conformation. However, there are several factors that can give rise to protein misfolding, such as gene mutations, errors made during translation and/or environmental stresses. Misfolded proteins are not functional and are often even cytotoxic. Therefore, cells have developed multiple strategies to prevent or to cope with proteotoxic stress.

Initially, chaperones such as heat shock proteins can help unstable or misfolded proteins to fold properly. If this process proves unsuccessful, misfolded proteins can be marked for degradation. The main route of protein degradation is via the ubiquitin-proteasome system (UPS). When this system fails or becomes saturated, the misfolded proteins will be concentrated into aggregates (Park et al., 2017). These small aggregates are then transported to the perinuclear region, where they form large aggresomes that can be degraded by lysosome-mediated autophagy.

Neurons are particularly reliant on these processes due to their post-mitotic nature, their distinctive morphology and their need to remain plastic (Tai and Schuman, 2008). Abnormal protein aggregation is a central hallmark and a widely recognized pathogenic mechanism in neurodegenerative disorders, including CMT (Lansbury and Lashuel, 2006). This common observation is highly suggestive for a defective

protein clearance machinery. Multiple studies have shown that mutations in several CMT2-associated proteins cause aggregate formation. It is particularly evident that mutations in *NEFL*, a component of the neurofilaments in the axon, can disrupt neurofilament assembly and can cause neurofilament aggregation (Brownlee et al., 2002; Perez-Olle et al., 2005, 2004, 2002; Sasaki et al., 2006). Additionally, mutations in *HSPB1* were shown to disrupt neurofilament assembly and lead to the formation of neurofilament- and HSPB1-containing aggregates (Zhai et al., 2007). Neurofilament aggregation is most clearly observed in giant axonal neuropathy, a childhood-onset neurodegenerative disease that is considered to constitute a continuum with CMT. This devastating disorder is caused by mutations in *gigaxonin* (*GAN*) (Bomont et al., 2000; Yang et al., 2007), which was shown to be a key player in UPS-mediated degradation of neurofilaments (Israeli et al., 2016; Mahammad et al., 2013). Altogether, these studies provide evidence for a contribution of defective protein turnover and protein aggregation in CMT2 pathogenesis. It should however be noted that protein aggregation might not be relevant for all forms of axonal CMT. In mutant *GARS* mice, for example, no ubiquitin accumulation was detected, indicating an absence of misfolded proteins (Stum et al., 2011).

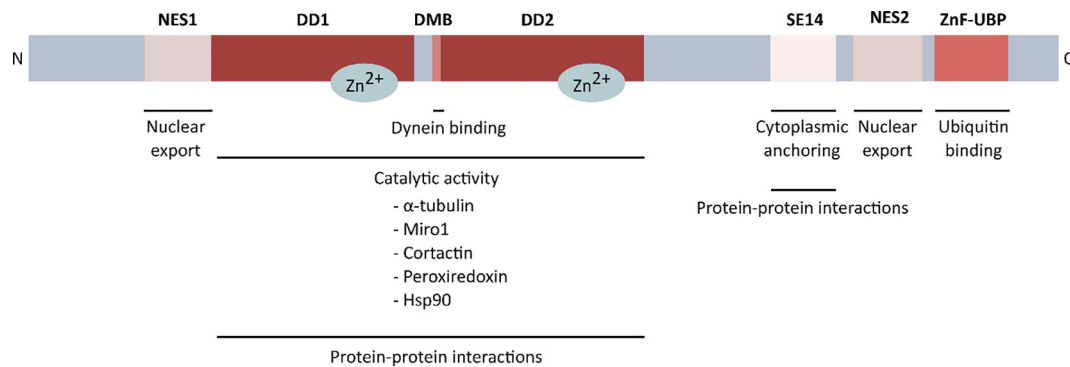


Fig. 1. The structural and functional domains of HDAC6. HDAC6 possesses two nuclear export signals, NES1 and NES2, that exclude the protein from the nucleus. In addition, it has a SE14 domain that stably anchors the protein in the cytoplasm and with which it can interact with partner proteins. Further, it contains two catalytic deacetylase domains, DD1 and DD2, with which HDAC6 deacetylates multiple substrates, including α -tubulin, Miro1, cortactin, peroxiredoxin and heat shock protein 90 (Hsp90). These deacetylase domains were also shown to be interaction sites for different partner proteins. A small DMB domain between both catalytic domains enables HDAC6 to bind dynein motor proteins, which execute retrograde transport along microtubule tracks. Last, a zinc finger ubiquitin-binding motif allows HDAC6 to bind ubiquitinated proteins.

3. HDAC6 is an unusual member of the HDAC family

Histone deacetylase 6 (HDAC6) is a member of the HDAC family of enzymes that are highly conserved amongst species, reaching back to prokaryotes (Haberland et al., 2009). Most if not all HDACs are critical regulators of transcription, either by epigenetically modifying histones or by regulating the activity of transcription factors (Delcuve et al., 2012).

HDAC6 is a rather unusual member of the HDAC family that executes peculiar functions. It possesses multiple exclusive structural features that have created unique properties compared to the other HDACs (Fig. 1). First, HDAC6 contains two deacetylation domains (DD1 and DD2) that are thought to have arisen from a gene duplication during evolution (Boyault et al., 2007). It is believed that their relative contribution to the deacetylase activity of HDAC6 is substrate-specific (Grozinger et al., 1999; Haggarty et al., 2003; Zhang et al., 2006, 2003; Zou et al., 2006). Moreover, the spatial arrangement between both domains has been shown to be crucial for the deacetylase activity of HDAC6 (Zhang et al., 2006). Second, HDAC6 possesses a zinc finger ubiquitin-binding motif (ZnF-UBP), which is located at the C-terminus. This enables HDAC6 to function in cell response pathways to proteotoxic stresses. It is remarkable that the ZnF-UBP domain of HDAC6 possesses the highest known affinity for ubiquitinated proteins, suggesting a crucial role of HDAC6 in protein clearance (Boyault et al., 2007). Importantly, it was shown that the binding of ubiquitinated proteins does not affect the coinciding deacetylase activity of HDAC6 (Seigneurin-Berny et al., 2001). Third, HDAC6 contains two leucine-rich nuclear export sequences (NES1 and NES2) that drive its localization to the cytoplasm. Furthermore, it contains a Ser-Glu tetradecapeptide repeating domain (SE14) which serves to stably retain HDAC6 in the cytoplasm (Bertos et al., 2004). Finally, a small domain between both catalytic domains enables HDAC6 to directly interact with dynein motor proteins (Kawaguchi et al., 2003). This domain is referred to as the dynein motor binding domain (DMB).

The immediate consequence of the combination of all these structural features is that it allows HDAC6 to function in various cytoplasmic processes that are both dependent and independent on its catalytic activity. Several targets for its deacetylase activity have been identified so far. These include α -tubulin, heat shock protein 90 (HSP90), peroxiredoxin, Miro1 and cortactin. The deacetylation of these proteins by HDAC6 regulates their activity and ultimately also the processes in which they are involved. At the same time, the ubiquitin binding motif of HDAC6 enables it to bind misfolded proteins that are marked for degradation. In combination with the capacity of HDAC6 to directly interact with dynein motor proteins, this suggests a role for HDAC6 as

an adaptor for cargos during intracellular transport (Kawaguchi et al., 2003).

Interestingly, it is becoming increasingly evident that HDAC6 can also interact with various proteins that are not prone to its deacetylase activity (Supplemental table 1). These interactors can reversibly, spatially and temporally regulate HDAC6 functions by modulating its catalytic activity. This is achieved either by physical interactions (Azuma et al., 2009; Benoy et al., 2018; Deakin and Turner, 2014; Ding et al., 2008; Mo et al., 2018; Perez et al., 2009; Salemi et al., 2017; Tokési et al., 2010; Wickström et al., 2010; Yan et al., 2013) or by kinase-mediated phosphorylation (Brush et al., 2004; Chen et al., 2010; Deribe et al., 2009; Du et al., 2015; Lafarga et al., 2012; Watabe and Nakaki, 2011; Williams et al., 2013). For the latter, it is the phosphorylation site that determines whether HDAC6 is stimulated or inhibited. Phosphorylation of serine 1035 by extracellular signal-regulate kinase (ERK) was shown to induce HDAC6 deacetylase activity, whereas phosphorylation of tyrosine 570 by epidermal growth factor receptor (EGFR) inhibited its activity (Deribe et al., 2009; Williams et al., 2013) (Supplemental table 1). In case of physical interactions, it seems that the responsible HDAC6 domain determines the net impact of the interaction. For example, tau was found to interact with HDAC6 via its SE14 domain, thereby decreasing HDAC6 activity (Ding et al., 2008; Perez et al., 2009). In contrast, HDAC6 can be activated by the estrogen receptor α (ER α), G-protein coupled receptor kinase 2 (GRK2) and mutant GARS, which all interact with its deacetylase domains (Azuma et al., 2009; Lafarga et al., 2012; Mo et al., 2018) (Supplemental table 1). The growing list of interacting proteins that regulate HDAC6 activity could provide a rationale on how HDAC6 can coordinate diverse cellular processes.

4. The noncanonical functions of HDAC6 in the nervous system

The unique properties of HDAC6 are well exemplified in neuronal cells, in which this protein meticulously regulates a variety of fundamental processes (Fig. 2). First, HDAC6 is a key regulator of axonal transport. The deacetylating substrates that are involved in this process are α -tubulin and Miro1. Second, HDAC6 restrains axonal outgrowth in adult neurons. By deacetylating several substrates that orchestrate axonal growth, including α -tubulin, Miro1 and cortactin, HDAC6 inhibits their specific functions and thus overall affects this process. Finally, HDAC6 plays a decisive role in the clearance of misfolded proteins. This is accomplished by the deacetylation of proteotoxic response proteins, the binding of ubiquitinated proteins and the interaction with dynein motor proteins. In this section, we briefly discuss the key role of HDAC6 in these three neuronal processes.

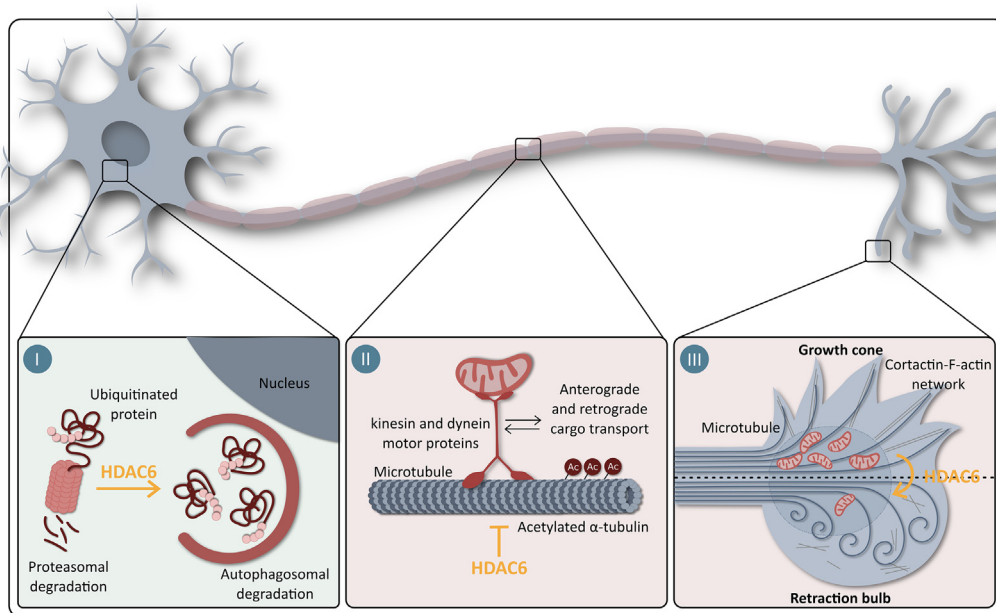


Fig. 2. Schematic overview of the different HDAC6 functions in CMT2 pathomechanisms. (I) HDAC6 mediates the shift from proteasomal- to autophagy-mediated protein degradation. Proteasomal degradation is the main route for protein degradation. In case the proteasomal degradation pathway is impaired or saturated, HDAC6 will redirect ubiquitinated proteins to the perinuclear region, where it will promote aggresome formation. These aggresomes can be cleared by autophagy. HDAC6 therefore plays a neuroprotective role by ensuring the degradation of toxic misfolded proteins. (II) HDAC6 negatively regulates anterograde and retrograde transport of cargoes, such as mitochondria, lysosomes, RNA, lipids and proteins, along the axons. This is mainly achieved by deacetylating α -tubulin, which is a component of microtubule tracks. The deacetylation of α -tubulin decreases the affinity of motor proteins to the microtubules, thereby slowing down axonal transport. HDAC6 therefore plays a neurotoxic role by repressing proper transport of essential cargoes along the axons. (III) HDAC6 is a critical mediator of pathways that restrain axonal outgrowth after injury, by the induction of retraction bulbs. These retraction bulbs are defined by destabilized microtubules, a disrupted cortactin-F-actin network and a reduced number of functional mitochondria. HDAC6 plays crucial roles in these three characteristics. First, it represses axonal transport of mitochondria to the nerve terminals by deacetylating α -tubulin and Miro1; second, it destabilizes microtubules by deacetylating α -tubulin; and third, it disrupts the cortactin-F-actin network by deacetylating cortactin. HDAC6 therefore plays a neurotoxic role by impeding axonal regeneration after axonal damage.

4.1. HDAC6 is a negative regulator of axonal transport

HDAC6 inhibition was shown to increase axonal transport of diverse cargoes in multiple neuronal types, including hippocampal neurons, motor neurons and sensory DRG neurons (Chen et al., 2010; d'Ydewalle et al., 2011; Kim et al., 2016). This implicates a critical regulatory role of HDAC6 in axonal transport, both in the CNS and the PNS.

The effects of HDAC6 on axonal transport have mainly been ascribed to its deacetylase activity on α -tubulin (Fig. 2). Alpha-tubulin is a major deacetylase substrate of HDAC6 and is a component of the cytoskeletal microtubule complex (Zhang et al., 2003). Acetylation of α -tubulin occurs on lysine residue 40 and is regulated by the opposing activities of α -tubulin acetyltransferase 1 (α TAT1) and HDAC6, or to a lesser extent SIRT2 (Hubbert et al., 2002; North et al., 2003).

Acetylated α -tubulin is most abundant in stable, long-lived microtubules, such as cilia and axons (Garnham and Roll-Mecak, 2012). The acetyl-modification ensures flexibility and protects the microtubules from damage (Portran et al., 2017). In a neuronal context, the acetylation level of α -tubulin is a critical denominator of axonal transport by improving the docking of motor proteins to microtubules (Dompierre et al., 2007; Reed et al., 2006).

The exact purpose of reversible acetylation in the axon during physiological conditions is not completely understood. A possible explanation is that spatial and temporal tuning of transport efficiency is needed to respond abruptly to cellular needs. Alternatively, switchable acetylation might be one of the factors that determines the net direction of transport, serving as some sort of code for motor proteins. Evidence for this theory is provided by the fact that acetylated α -tubulin preferentially binds kinesins, the motor proteins that effectuate anterograde cargo transport (Friedman et al., 2010; Reed et al., 2006).

Recently, Miro1 was found to be an additional important HDAC6

substrate contributing to its regulation of axonal transport, and more specifically that of mitochondria (Kalinski et al., 2019; Van Den Bosch, 2019). Miro1 is a calcium-binding outer mitochondrial membrane protein and its acetylation on lysine 105 was shown to increase mitochondrial axonal transport (Kalinski et al., 2019; Van Den Bosch, 2019).

4.2. HDAC6 counteracts axonal outgrowth

In addition, HDAC6 is a principal regulator of axonal outgrowth after injury (Fig. 2). There are three ways in which HDAC6 interferes with this delicate process: (1) its effect on axonal transport, (2) on microtubular stabilization and (3) on actin-dependent protrusion.

As a principal regulator of axonal transport, it controls the provision of essential components at the growth cone. The delivery of functional mitochondria, lipids and proteins are crucial for axonal outgrowth. As discussed above, HDAC6 negatively regulates axonal transport by deacetylating α -tubulin and Miro1 (Kalinski et al., 2019; Zhang et al., 2008, 2003). Accordingly, the deacetylation of these substrates also impedes axonal outgrowth (Kalinski et al., 2019; Riviccio et al., 2009; Tapia et al., 2010).

A second mode of action of HDAC6 in regulating axonal outgrowth is also mediated by α -tubulin deacetylation. The deacetylation of α -tubulin was shown to destabilize microtubules, which provokes axon degeneration and opposes axon regeneration in the CNS (Ertürk et al., 2007; Hellal et al., 2011; Ruschel et al., 2015; Wong et al., 2018).

In addition to its role in microtubule-dependent axonal outgrowth, HDAC6 influences actin-dependent outgrowth by disrupting the cortactin-F-actin network (Zhang et al., 2007). Cortactin interacts with F-actin, thereby regulating its polymerization (Wu and Parsons, 1993). At the growth cone, F-actin polymerization drives protrusion that propel

the leading edge (Dent et al., 2011). The deacetylation of cortactin by HDAC6 alters its ability to bind F-actin, thereby halting membrane protrusion and thus axonal outgrowth (Zhang et al., 2007).

4.3. HDAC6 plays a central role in protein aggregate elimination

HDAC6 is a key player in various cellular response mechanisms to deleterious misfolded proteins (Fig. 2). Misfolded proteins are typically marked with ubiquitin for elimination. In case the main UPS-mediated degradation system becomes saturated or is impaired, ubiquitinated misfolded proteins are transported to the perinuclear region, where all misfolded proteins are stored in aggresomes for removal via lysosome-dependent autophagy.

HDAC6 is a crucial player in this shift from UPS- to autophagy-mediated degradation and it was suggested that it acts as sensor of proteasome inhibition (Jiang et al., 2008). HDAC6 has a ubiquitin-binding domain with the highest known affinity for ubiquitinated proteins (Boyault et al., 2007). When ubiquitinated proteins are present in the cell, they will interact with HDAC6. In case the proteasome is functional, a complex consisting of valosin-containing protein (VCP)/p97 interacts with HDAC6, causing the dissociation of the HDAC6/misfolded protein complexes so that protein degradation via the main proteasomal route can occur (Pernet et al., 2014). However, when the proteasomal degradation pathway is impaired, the interaction of the ubiquitinated protein with HDAC6 will remain. The fate of the misfolded proteins thus partially depends on the balance between HDAC6 and VCP/p97.

For transporting the ubiquitinated protein to the perinuclear region in the cell body, HDAC6 functions as a molecular adaptor that links the misfolded protein to dynein, the motor protein that is responsible for retrograde transport. Therefore, HDAC6 links protein degradation and transport both structurally and functionally, mediating the protective clearance of toxic misfolded proteins. In the perinuclear region, HDAC6 can interact with FAT10, which promotes FAT10-mediated aggresome formation (Kalveram et al., 2008).

Apart from its role in mediating the shift from proteasomal- to autophagy-mediated protein degradation, HDAC6 is involved in preventing further formation of misfolded proteins. It does so by activating heat shock proteins in multiple ways. Normally, HDAC6 is bound to the heat shock protein 90/heat shock factor 1 (HSP90/HSF1) complex (Boyault et al., 2007). When HDAC6 binds ubiquitinated proteins, it dissociates from this complex, thereby releasing HSP90 and HSF1 (Boyault et al., 2007). HSF1 is a transcription factor that can induce the expression of heat shock protein 70 (HSP70) and heat shock protein 27 (HSP27), which act as chaperones. These chaperones, in combination with the released and activated (acetylated) HSP90, can prevent further formation of misfolded proteins.

In conclusion, most functions of HDAC6 in the clearance of misfolded proteins are mediated independent of its deacetylation activity, with the exception of HSP90 deacetylation. HSP90 activity is regulated on different levels: it works together with a large group of co-chaperones and is subjected to both phosphorylation and acetylation at multiple sites (Buchner and Li, 2013). The specific role and consequences of HSP90 deacetylation by HDAC6 remains to be investigated.

5. HDAC6: An emerging player in axonal CMT?

Interestingly, HDAC6 fulfills a key role in many CMT2-related processes. As mentioned in the first section of this review, defects in axonal transport and the subsequent inability of axonal regeneration can be directly linked to the functions of several CMT2-related genes. However, there is no obvious direct link with axonal transport for certain CMT2-related genes. In these cases, HDAC6 could be the missing link between the CMT2 genes and the axonal transport phenotype.

We propose a model in which several mutant CMT2-associated proteins serve as co-factors that aberrantly activate HDAC6, thereby

altering the acetylation status of various HDAC6 substrates and finally also the critical processes of axonal transport and axonal regeneration.

Most evidence comes from two recent studies that showed that mutant GARS, the glycyl tRNA synthetase, aberrantly interacts with HDAC6 (Benoy et al., 2018; Mo et al., 2018). Both studies demonstrated that α -tubulin acetylation was significantly decreased in mutant GARS mouse models (Benoy et al., 2018; Mo et al., 2018), implying an imbalance in the α -TAT1-HDAC6 enzymatic machinery. Interestingly, the study of Mo et al. revealed elevated HDAC6 activity in cells transfected with mutant GARS (Mo et al., 2018), suggesting a gain-of-toxic function of HDAC6 by the aberrant interaction. It is reasonable to assume that other CMT2-related tRNA synthetases like YARS, MARS, AARS and HARS (Laurá et al., 2019), which are structurally and functionally similar to GARS, also interact with HDAC6. In fact, it was shown that CMT2-linked mutations in HARS and YARS induce a conformational change, thereby exposing a new surface for potential pathological interactions (Blocquel et al., 2019, 2017). Also in case of *HSPB1*, a similar mechanism involving HDAC6 could be likely, as a decrease in α -tubulin acetylation was observed in the affected tissues of mutant *HSPB1* mice (d'Ydewalle et al., 2011) and as it has been shown to interact with HDAC6 (Gibert et al., 2012). Future studies should reveal whether CMT2-related mutations in *HSPB1* indeed affect this interaction.

In conclusion, there are many indications that HDAC6 actively participates in several forms of CMT2 pathogenesis. However, the role of HDAC6 in CMT has been mainly deduced from effects observed after inhibition of its enzymatic activity, which is discussed in more detail in the following section.

6. Targeting HDAC6 to combat CMT

HDACs are druggable targets and compounds that inhibit this class of enzymes have been purified from natural sources, or have been synthesized. Class I, II and IV HDACs require a zinc ion for their deacetylation activity and share conserved residues in their active site. HDAC inhibition is achieved by compounds that can bind the zinc ions and surrounding residues with a zinc-binding group (Bradner et al., 2010). This binding determines the inhibitory potency of the drug. A linker spans the narrow channel of the HDAC enzymes and connects the zinc-binding group to a capping structure (Bradner et al., 2010). The cap element of the compound can bind to residues at the rim-region of the active site cavity, contributing to the selectivity toward the different HDAC isoforms (Bradner et al., 2010).

Due to their role in several diseases, HDACs have emerged as promising drug targets. Targeting HDAC6 is particularly attractive, as HDAC6 knockout mice remain viable and show no obvious phenotype (Zhang et al., 2008). Due to its unique structural features, HDAC6 has distinct functions compared to the other HDACs. It is therefore unlikely that other HDACs would compensate for HDAC6 in the HDAC6 knockout mice. Moreover, the dimensions of the catalytic rim region of HDAC6 differ from that of the other HDACs, enabling the design of selective inhibitors (Butler et al., 2010). The first selective HDAC6 inhibitor, tubacin, showed good selectivity, but its high lipophilicity and extensive synthesis process made it a poor drug candidate (Haggarty et al., 2003). Since then, lots of efforts have been made to develop selective HDAC6 inhibitors with improved pharmacokinetic properties and low toxicity, including tubastatin A (Butler et al., 2010; De Vreese et al., 2017).

6.1. The efficacy of HDAC6 inhibitors in preclinical models of CMT2

We tested the therapeutic efficacy of HDAC6 inhibitors in the context of axonal CMT. Treatment of mutant *HSPB1*^{S135F} mice with tubastatin A reversed the motor and sensory deficits and increased muscle reinnervation, which was associated with restoration of α -tubulin acetylation levels in the peripheral nerves (d'Ydewalle et al., 2011). Moreover, tubastatin A treatment of cultured DRG neurons isolated

from mutant HSPB1^{S135F} mice rescued the axonal transport defects of mitochondria, providing a mechanistic explanation for the behavioral outcomes (Table 1) (d'Ydewalle et al., 2011).

Recently, we and others discovered that the axonal transport phenotype and the therapeutic efficacy of HDAC6 inhibitors could be extended to GARS-mediated CMT2 (Table 1) (Benoy et al., 2018; Mo et al., 2018). In DRG cultures from two mutant GARS mouse models, GARS^{C201R} and GARS^{P234KY}, axonal transport defects were observed. Intriguingly, not only mitochondrial transport but also nerve growth factor (NGF) transport was hampered (Mo et al., 2018). Moreover, the defects in axonal transport were already noted at a pre-symptomatic age, suggesting a causal link with disease pathogenesis (Mo et al., 2018). Treatment with tubastatin A rescued the transport defects in the DRG neurons and reversed the behavioral and histopathological phenotypes of both models.

The *in vitro* and *in vivo* HSPB1^{S135F} phenotypes were subsequently used to screen for new HDAC6 selective compounds with improved pharmacokinetic properties in the search for a potential new therapy for CMT2 (Benoy et al., 2017; Shen et al., 2016). Moreover, iPSC-derived motor neurons from mutant HSPB1^{S135F} patients showed a similar transport phenotype and proved useful as a screening platform to test novel selective HDAC6 inhibitors with improved drug-like properties (Kim et al., 2016).

Overall, research in animal and cellular models strongly indicates that HDAC6 inhibition is a promising therapeutic strategy for at least two different genetic causes of axonal CMT. As axonal transport defects are a common hallmark in various CMT2 genes (Table 1), HDAC6 could also be an important therapeutic target for other genetic forms of CMT2.

6.2. Possible modes of action

Based on multiple studies, it is now clear that HDAC6 inhibitors have neuroprotective properties in several preclinical models of CMT2. Through hyperacetylation of various targets, HDAC6 inhibitors can affect multiple pathways and induce a wide range of effects. We suggest three major molecular mechanisms of action based on: (1) their stimulatory effects on axonal transport, (2) their effects on axonal outgrowth and (3) their competition with aberrantly interacting proteins.

It is widely appreciated that HDAC6 inhibitors improve axonal transport in a number of neurodegenerative models (Prior et al., 2018; Simões-Pires et al., 2013). The efficiency of axonal transport is controlled by the levels of α -tubulin acetylation. The deacetylation reaction, restraining axonal transport, is executed by HDAC6 (Zhang et al., 2003). Transport along the axon is especially crucial for peripheral neurons that must move cargo over very long distances. Several studies showed a significant reduction in acetylated α -tubulin levels in CMT2 models, which was coupled with axonal transport defects (Benoy et al., 2018; d'Ydewalle et al., 2011; Kim et al., 2016; Mo et al., 2018). The selective inhibition of HDAC6 could restore α -tubulin acetylation levels and improve axonal transport (Benoy et al., 2018; d'Ydewalle et al., 2011; Kim et al., 2016; Mo et al., 2018; Adalbert et al., 2020). In animal models for CMT2, this therapeutic approach exerted beneficial effects on the disease phenotype (Benoy et al., 2018; d'Ydewalle et al., 2011; Mo et al., 2018).

In addition to its effect on axonal transport, it was shown that HDAC6 inhibitors can protect against axonal damage (Butler et al., 2010; Kalinski et al., 2019; Riviello et al., 2009). Under basal conditions, HDAC6 inhibitors do not affect neurite outgrowth, but increase growth cone size and induce mitochondrial accumulation in the nerve endings (Kalinski et al., 2019; Riviello et al., 2009). A recent study by Kalinski et al. showed that these responses can protect DRG neurons against axonal damage (Kalinski et al., 2019). Oxidative stress-induced damage typically induces growth cone retraction (Butler et al., 2010; Kalinski et al., 2019; Riviello et al., 2009). However, pretreatment with HDAC6 inhibitors prevented growth cone collapse, thereby

impeding axonal degeneration (Butler et al., 2010; Kalinski et al., 2019).

Several studies showed that also symptomatic treatment with HDAC6 inhibitors can stimulate axonal outgrowth after injury (Kalinski et al., 2019; Riviello et al., 2009). HDAC6 is a major negative regulator of axonal outgrowth in a nonpermissive growth environment reminiscent to the injured CNS. To simulate this environment, neurons can be exposed to myelin-associated glycoprotein (MAG) or chondroitin sulfate proteoglycan (CSPG), which typically leads to growth cone collapse and halts neurite outgrowth (McKerracher and Rosen, 2015; Siebert et al., 2014). It was shown that treatment of DRG cultures with selective HDAC6 inhibitors enabled neurite outgrowth, irrespective of the repulsive guidance cues (Kalinski et al., 2019; Riviello et al., 2009). Interestingly, Riviello et al. demonstrated that treatment on the axon alone was sufficient to mediate this effect, excluding a transcriptional-dependent response (Riviello et al., 2009). However, it should be noted that the PNS generally shows a more permissive environment for nerve regeneration after injury. Degenerating peripheral neurons in CMT2 patients are therefore presumably not exposed to the same inhibitory cues. Nevertheless, animal studies indicate that symptomatic treatment of CMT2 mice with HDAC6 inhibitors can stimulate regeneration of damaged peripheral neurons, as observed by increased muscle innervation and improved electrophysiological measurements (Benoy et al., 2018; d'Ydewalle et al., 2011; Mo et al., 2018).

Another, quite distinct mechanism that could contribute to the therapeutic effects of HDAC6 inhibitors observed in mutant HSPB1 and GARS models is that the small-molecule HDAC6 inhibitors go in competition with certain interaction partners of HDAC6. Two studies showed that mutant GARS aberrantly interacts with HDAC6 via its second deacetylation domain, thereby increasing HDAC6 activity (Benoy et al., 2018; Mo et al., 2018) (Supplemental table 1). HDAC6 inhibitors also primarily bind to this second catalytic domain (Haggarty et al., 2003). It is therefore reasonable to hypothesize that treatment with HDAC6 inhibitors disrupts aberrant protein-protein interactions, thereby restoring HDAC6 activity and ultimately also the acetylation balance of its substrates. Evidence for this theory is provided by the study of Benoy et al., demonstrating a disrupted GARS-HDAC6 interaction upon tubastatin A treatment (Benoy et al., 2018) (Supplemental table 1).

7. Concluding remarks

A compelling body of evidence has established disturbances in cytoskeletal proteins and axonal transport as a common mechanism underlying axonal CMT. In certain genetic forms of CMT2 there is a clear, direct link with the axonal transport machinery. In other cases, HDAC6 could be the missing link connecting the underlying gene with this process.

In addition to its repressive effects on axonal transport, HDAC6 is a critical mediator of signaling pathways that restrain regeneration after axonal damage. Besides these potential neurotoxic functions, HDAC6 actively contributes to cellular defense mechanisms against proteotoxicity by means of its ubiquitin-binding capacities (d'Ydewalle et al., 2012). Various studies have provided evidence for defects in these three processes in axonal CMT, highlighting HDAC6 as a potential therapeutic target. Interestingly, the neurotoxic properties of HDAC6 are dependent on its deacetylase activity, whereas most of its neuroprotective functions are ubiquitin-dependent (d'Ydewalle et al., 2012). Small-molecule HDAC6 inhibitors that only target its deacetylase activity are therefore a good therapeutic strategy to cease the neurotoxic effects of HDAC6, while leaving its protective properties intact. In fact, the efficiency of this strategy was already demonstrated by several preclinical studies that tested selective HDAC6 inhibitors in CMT2 disease models.

Decreased levels of acetylated α -tubulin and axonal transport defects have been observed in other neurodegenerative disorders as well

(Guo et al., 2019). This suggests that HDAC6 inhibitors could offer therapeutic perspectives for a wide spectrum of neurodegenerative diseases. Multiple studies indeed demonstrated favorable effects of HDAC6 inhibitors in models of acquired neuropathies (Van Helleputte et al., 2018), Alzheimer's disease (Yao et al., 2014), Parkinson's disease (Godena et al., 2014) and amyotrophic lateral sclerosis (Guo et al., 2017). Future studies will have to uncover whether HDAC6 inhibitors can counteract the symptoms in the demyelinating form of CMT (CMT1).

A phase 1 clinical trial intended to evaluate the safety and anti-tumor activity of HDAC6 inhibitors showed a good tolerability in humans (Yee et al., 2016). Moreover, there are currently multiple clinical trials ongoing testing HDAC6 inhibitors in patients with different types of cancer as well as for other diseases. For an overview of the recruiting, ongoing and completed clinical trials, we refer to a recent review (Shen and Kozikowski, 2020).

Repurposing these HDAC6 inhibitors for CMT2 would be an interesting opportunity. However, there might be some limitations to HDAC6 inhibitor therapy. Although it has been shown that conventional HDAC6 knockout mice remain viable and show no obvious phenotype, a recent study demonstrated an impaired inflammatory response in these mice (Moreno-Gonzalo et al., 2017). As a consequence, one should be cautious for long-term systemic exposure of patients to these drugs. An intermittent treatment schedule could be considered to circumvent this issue.

In conclusion, there are strong indications that axonal transport defects play an active role in CMT2 pathogenesis. Moreover, defects in axonal regeneration and misfolded protein handling are observed. As HDAC6 plays an essential role in all these processes, a rational therapeutic strategy would be the use of selective HDAC6 inhibitors. Overall, we conclude that HDAC6 inhibitors are a promising therapeutic strategy in axonal CMT.

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Appendix A. Supplementary data

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